Appendix: Reciprocal relationships, reverse causality, and temporal ordering: Testing theories with cross-lagged panel models

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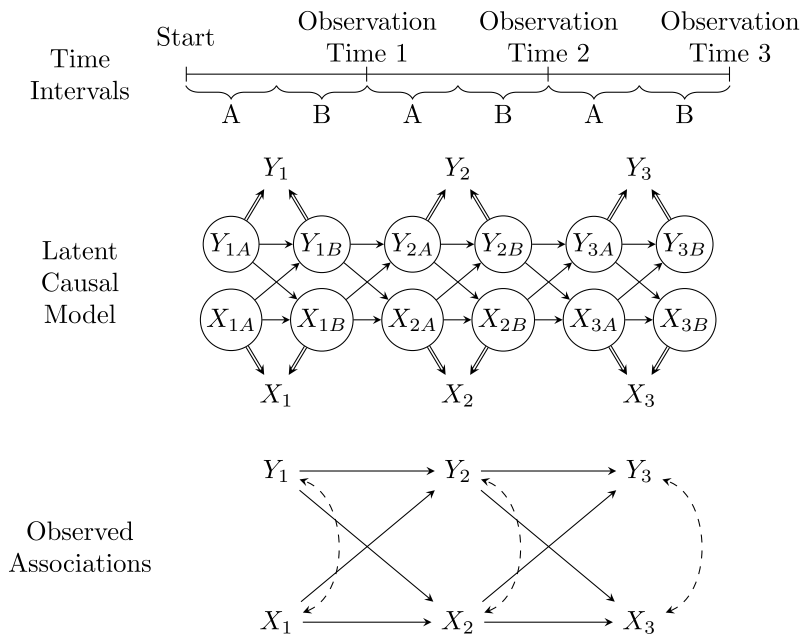

Figure 1: Reciprocality due to temporal composite variable.

Figure 1 illustrates a situation common in panel survey designs where the timing of data collection does not align with the timing of the causal process under investigation and the focal variables are cumulative measures over an interval of time, e.g., number of offenses committed. Here the recall period for data collection encompasses 1 unit of time (e.g., an entire year) but the causal process operates over smaller intervals (e.g., six months) denoted A and B. This is represented by cross-lagged effects between latent X and Y variables occurring across each smaller interval. Because data were only collected at each observation time, the measured values represent some composite of the values from both intervals A and B. For example, if Y_1 is the number of self-reported criminal offenses in a year, it would be the simple sum of offenses $Y_{1A} + Y_1 B$. While we may know this sum, without additional information, we are uncertain of the value of either Y_{1A} or $Y_1 B$ (unless $Y_1 = 0$). Borrowing the notation of Berrie et al. (@025), the double-arrows indicate that Y_t and X_t are *deterministic* variables, i.e., they are fully determined by *some* combination of the underlying latent variables, even if we do not know their relative contribution[1].

Note that despite $X_t = X_{ta} + X_{tb}$, X_t is not compositional data in the sense of Berrie et al. (2025), because the parents X_{ta} and X_{tb} are not simultaneously determined. Rather, X_t is a composite derived variable where the parents are unobservable or latent.

Additional issues:

[1]If Y_{tA} and/or Y_{tB} contain measurement error, this full measurement error is propagated into the composite variable as well